



Sustainable Fluvial Mobility and Regional Connectivity on the Banjarmasin–Muara Teweh Corridor, Indonesia: Factor-Analytic Evidence from Water-Bus Users

Nevy Farista Aristin¹, Muhammad Muharram Azhari¹, Karunia Puji Hastuti¹, Agus Purnomo^{2*},
Deasy Arisanty¹, A. Riyan Rahman Hakiki¹, Sidharta Adyatma¹

¹ Department of Geography Education, Universitas Lambung Mangkurat, 70123 Banjarmasin, Indonesia

² Department of Social Studies, Universitas Negeri Malang, 65145 Malang, Indonesia

* Correspondence: Agus Purnomo (agus.purnomo.fis@um.ac.id)

Received: 07-31-2025

Revised: 09-19-2025

Accepted: 01-28-2026

Citation: N. F. Aristin, M. M. Azhari, K. P. Hastuti, A. Purnomo, D. Arisanty, A. R. R. Hakiki, and S. Adyatma, “Sustainable fluvial mobility and regional connectivity on the Banjarmasin–Muara Teweh Corridor, Indonesia: Factor-analytic evidence from water-bus users,” *Int. J. Transp. Dev. Integr.*, vol. 10, no. 2, pp. 478–491, 2026. <https://doi.org/10.56578/ijtdi100210>.



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Abstract: River transport remains central to mobility in Banjarmasin, Indonesia, where riverine settlements and uneven road access continue to shape everyday travel and regional connectivity. This study examines the factors associated with water-bus use on the Banjarmasin–Muara Teweh corridor and evaluates how these factors relate to sustainable fluvial mobility. A quantitative survey was conducted with 60 passengers of the Pancar Mas water bus at Banjar Raya Pier, supported by brief interviews on travel motives. The questionnaire covered economic, regional, and social indicators. Data adequacy was tested using the Kaiser-Meyer-Olkin (KMO) measure and Bartlett’s test, followed by principal component extraction, Varimax rotation, and confirmatory factor analysis (CFA). The results showed acceptable sampling adequacy ($KMO = 0.614$) and significant inter-variable correlations (Bartlett’s test, $p < 0.001$). Three factors were retained and together explained 69.55% of the variance. Economic conditions formed the strongest factor, with income opportunity (loading = 0.879), occupation type (0.800), and job availability (0.735) as the main indicators. Regional characteristics were represented by transport availability (0.913) and accessibility (0.838), while the social dimension was reflected in housing ownership status (0.851). The CFA results also showed acceptable model fit, with $\chi^2/df = 2.15$, goodness of fit index (GFI) = 0.91, comparative fit index (CFI) = 0.93, and root mean square error of approximation (RMSEA) = 0.062. The findings indicate that water-bus use in this corridor is shaped by livelihood opportunities, transport access, and settlement security. The study provides empirical evidence for maintaining river transport as part of regional connectivity and sustainable transport planning in riverine areas.

Keywords: Fluvial transport; Water bus; Sustainable mobility; Transport accessibility; Regional connectivity; Confirmatory factor analysis; South Kalimantan

1 Introduction

Banjarmasin City, located in South Kalimantan, Indonesia, is famously known as the “City of a Thousand Rivers” due to its network of over a hundred rivers that traverse the urban landscape, making it one of the most complex river-based ecosystems in Southeast Asia [1, 2]. These rivers not only shape the city’s natural and ecological characteristics but also play a vital role in its transportation system [3]. Serving as primary corridors, the rivers connect areas that are geographically separated and difficult to access via land-based infrastructure [4]. As a result, river-based transportation, particularly water buses, has become the main mode of mobility for residents. Despite the rapid development of land infrastructure such as highways and public transit systems, in recent years, river transport continues to hold significant appeal for the people of Banjarmasin [5].

One of the most prominent river transportation routes is the Banjarmasin–Muara Teweh corridor, which serves as a vital alternative for communities residing in inland and inter-district border areas. Although land-based transportation infrastructure has significantly improved over the past decades, water buses or river vessels remain an

essential mode of transport due to their relatively low cost, superior geographic accessibility, and the historical and cultural values attached to this mode of mobility [6].

The continued reliance on river transportation amidst the rapid expansion of modern land-based transport raises a critical question: why do communities still prefer river-based modes as their primary means of mobility? The answer extends beyond mere practical considerations and reflects a complex interplay of economic, social, and geographic factors that shape mobility preferences. For populations living along riverbanks or in remote areas, water transport often remains the only efficient option for accessing essential services such as education, healthcare, and economic opportunities [7].

However, despite long-standing scholarly attention to human mobility within the field of transportation studies, most research remains concentrated on land-based modes and urbanization issues [8–10]. Studies that specifically investigate river-based mobility in the context of sustainability and access to remote areas are still limited. Previous works have predominantly emphasized economic aspects such as income levels or travel costs [11, 12], but have not adequately incorporated the influence of social dimensions (e.g., kinship ties, community attachment to waterways) and geographic variables (such as topographic constraints and accessibility in riverine regions). This imbalance leaves a conceptual gap in explaining why communities continue to rely on river transport despite rapid improvements in land-based infrastructure.

Therefore, this study is designed to fill the gap by systematically analyzing how economic, social, and geographic factors jointly influence community preferences for river transportation along the Banjarmasin–Muara Teweh corridor. Unlike prior studies that treat these determinants in isolation, this research employs a confirmatory factor analysis (CFA) framework to validate the interrelationships among the three dimensions, thereby offering a more integrated and empirically tested model of riverine mobility. By doing so, the study extends beyond earlier works and contributes a novel perspective that highlights the multidimensional and interdependent nature of sustainable river transport preferences.

2 Literature Review

2.1 River Transportation and Tropical Geography

River transportation has long served as a fundamental mobility system, particularly in tropical regions abundant in water resources, such as Kalimantan. Rivers function not only as transportation corridors but also as vital living spaces that support domestic activities, trade, and local culture [13]. Banjarmasin, known as the “City of a Thousand Rivers,” exemplifies how an extensive water network shapes urban morphology and serves as a primary channel for the movement of goods and people [14–16].

In the context of tropical geography, such as that of South Kalimantan, the terrain is largely dominated by swamps, dense forests, and major river systems, presenting significant challenges to the development of land infrastructure. As a result, river transport becomes a more geographically and economically rational mode of transportation [6, 17]. Rivers function as structural elements within the transportation network, capable of reaching inland regions that remain inaccessible via permanent road infrastructure [18, 19].

2.2 Theories of Population Mobility and Mode Choice

The concept of mobility in transportation studies refers to the ability of individuals or groups to move from one location to another in order to meet their daily needs [8]. Mobility is influenced by various internal factors (such as individual characteristics) and external factors (such as accessibility and infrastructure) [20]. The selection of a mode of transportation, whether land, water, or air, is driven by a combination of practical considerations (such as time, cost, and comfort) and sociocultural factors (such as habits and collective preferences) [21, 22].

Mode choice is often not purely economic but also emotional and symbolic. In this context, communities accustomed to living along riverbanks tend to prefer water-based transport, as it has become an integral part of their local identity [23]. This aligns with the concept of “habitual mobility,” which suggests that mode choice can be inherited across generations and shaped by community norms and cultural traditions [24, 25].

2.3 Transport Sustainability: Economic, Social, and Geographic Dimensions

Sustainability in transportation requires the integration of economic efficiency, social equity, and environmental preservation. A sustainable transport system should be able to reduce congestion, lower emissions, and improve accessibility for vulnerable and underserved communities [26, 27].

River transportation is generally considered more environmentally friendly than land-based modes, as it produces lower emissions and does not require extensive land clearing. Economically, the operational costs of riverboats are relatively low, making them a viable and affordable option, especially for communities in remote areas. From a social perspective, river transport supports the mobility of isolated populations, enabling access to essential public services such as education and healthcare [28, 29].

2.4 Academic Gaps in the Study of River Mobility

The academic literature on transportation in Indonesia still exhibits a strong bias toward the development of land-based modes. River transportation is often addressed only marginally in policy reports or infrastructure development studies. A deeper understanding of alternative transport modes is crucial for achieving more equitable and inclusive regional planning [30].

Research on river mobility also tends to concentrate on technical aspects, such as vessel types or port capacity, while largely overlooking sociocultural dimensions and user preferences. Therefore, a more comprehensive and interdisciplinary approach is needed to capture the complexity of river transportation systems, particularly within remote and community-based contexts [31].

2.5 Confirmatory Factor Analysis in Social Transportation Research

CFA is a statistical technique within the structural equation modeling (SEM) framework, used to test the validity of theoretical constructs based on empirical data [32]. CFA is particularly well-suited for this study as it enables researchers to examine the extent to which economic, social, and geographic factors influence individuals' decisions in choosing river transportation.

CFA also allows for the identification of the most significant indicators associated with each factor, and for assessing the overall model fit [33]. In the context of this research, CFA helps reveal the latent structure underlying preferences for river-based transportation modes and confirms whether the proposed theoretical framework is supported by field data [34–36].

3 Methods

This study adopts a quantitative approach aimed at measuring and analyzing the relationships among various factors that influence the choice of river transportation modes in Banjarmasin, particularly along the Banjarmasin–Muara Teweh route (Figure 1). The quantitative approach was selected because it enables the collection of numerical data that can be objectively analyzed and tested using valid statistical techniques [37]. Through this approach, the study investigates the influence of economic, social, and geographic factors on users' decisions to choose river transport, and examines the model linking these factors using robust statistical analysis methods.

3.1 Research Design

This study employs a quantitative descriptive research design using CFA. The design aims to systematically describe the variables influencing the selection of river transportation and to analyze the relationships among those variables. The descriptive component refers to how the study outlines the characteristics and influencing factors of river transport usage in the Banjarmasin–Muara Teweh area, while the analytical component focuses on identifying and measuring the strength of the relationships among the factors involved in transport decision-making.

The hypothesized factors in this study include economic factors (such as income and transportation costs), social factors (including social status and social networks), and geographic factors (such as accessibility to transportation points and travel duration). CFA is employed to confirm whether the factor structure proposed in the research model aligns with the collected data and to validate the strength of the relationships among the variables.

3.2 Population and Sample

The population in this study consists of passengers using the water bus service on the Banjarmasin–Muara Teweh route. This research employs an accidental sampling technique, in which samples are drawn randomly from the existing population based on the availability and willingness of respondents at the time of data collection. Accidental sampling was chosen because it allows the researcher to obtain data from respondents who are readily accessible and willing to participate, without the need for more complex randomization procedures. The sample for this study includes 60 randomly selected respondents, all of whom were passengers of the Pancar Mas water bus operating on the Banjarmasin–Muara Teweh route.

3.3 Research Instruments

The instrument used in this study is a closed-ended questionnaire consisting of a series of questions designed to measure the factors influencing the choice of transportation mode. The questionnaire was developed to collect information related to three main factors:

- Economic Factors: including income level, type of employment, and transportation costs.
- Social Factors: including social status, social networks, and proximity to family.
- Geographic Factors: including accessibility to transportation points, travel duration, and ease of access.

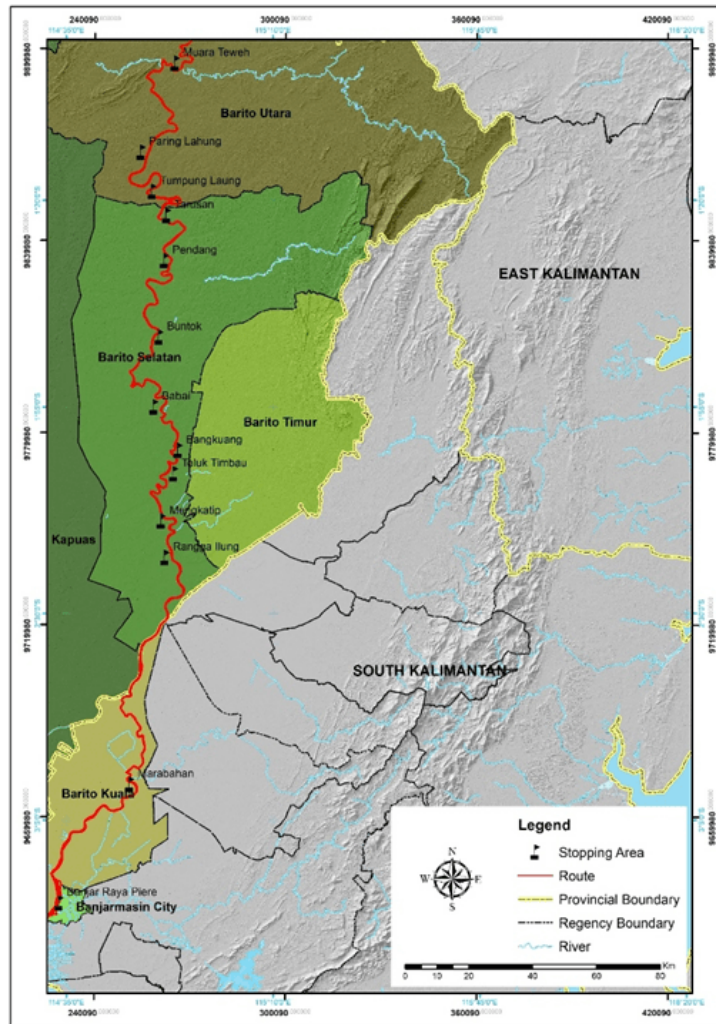


Figure 1. Banjarmasin–Muara Teweh Route

3.4 Data Collection

Data collection was conducted at the Banjar Raya Pier, which serves as the main departure point for the Pancar Mas water bus. Data were gathered over the course of one month, specifically every Monday, as the Pancar Mas water bus operates only once a week on that day. Questionnaires were distributed to passengers at the pier prior to departure. The questionnaire employed a 5-point Likert scale, in which respondents were asked to rate each statement from “Strongly Disagree” to “Strongly Agree.”

In addition to the questionnaire, supporting interviews were conducted to obtain deeper qualitative insights. These interviews were carried out with several respondents who voluntarily agreed to provide further explanation regarding their reasons for choosing river transportation. The interviews took place after the questionnaire had been completed and focused on the respondents’ experiences using river transport, as well as the key factors influencing their decisions, such as comfort, cost, and travel time.

3.5 Data Analysis

Data analysis was conducted to test the proposed hypotheses and to understand the relationships among the factors influencing the choice of river transportation modes. This study employed CFA to validate the hypothesized factor model, which consists of three primary dimensions: economic factors, social factors, and geographic factors. The data analysis process was carried out using IBM SPSS version 21.0 and AMOS (Analysis of Moment Structures) software.

3.5.1 Sample adequacy test

Before conducting CFA, it is essential to ensure that the collected data are sufficiently representative and meet the assumptions for analysis. Therefore, two preliminary tests were carried out: the Kaiser-Meyer-Olkin (KMO)

Measure and Bartlett's Test of Sphericity.

- The KMO Test is used to assess the adequacy of the sample for factor analysis. The KMO value ranges from 0 to 1, with higher values indicating better suitability for factor analysis. A KMO value above 0.5 is generally considered acceptable, with higher values suggesting stronger sampling adequacy.
- Bartlett's Test evaluates the null hypothesis that the variables are uncorrelated in the population (i.e., the correlation matrix is an identity matrix). A significant result ($p < 0.05$) indicates that the data are appropriate for factor analysis.

3.5.2 Factor extraction

In this stage, the Principal Component Analysis (PCA) technique was employed to extract the factors influencing the use of river transportation. PCA is used to identify the underlying factors that account for the greatest variability in the dataset. It reduces the dimensionality of the data into a smaller set of components that can explain observed patterns.

Each variable was assessed based on how much of its variance could be explained by the resulting components. Components with an eigenvalue greater than 1 were considered significant, indicating that they explain more variance than a single observed variable and were therefore retained as principal factors for the model.

3.5.3 Factor rotation

To facilitate the interpretation of factor analysis results, a Varimax rotation was applied. Rotation is a technique used to simplify the understanding of factor structure by organizing variables to load more strongly on one factor and less on others. Varimax rotation, an orthogonal rotation method, aims to maximize the squared variance of the factor loadings. This technique enhances interpretability by encouraging each variable to have a high loading on only one factor, making the structure more distinct and easier to explain [38].

The rotation process produces factor loadings that indicate the strength of the relationship between each variable and its corresponding factor. The higher the factor loading, the stronger the association between the variable and the factor.

3.5.4 Confirmatory factor analysis

After extracting and rotating the factors, the next step was to conduct CFA using AMOS software. CFA is used to confirm whether the resulting factor structure aligns with the proposed theoretical model [39]. It evaluates how well the collected data support the hypothesized model. Several indices are commonly used to assess model fit, including:

- Chi-Square/ df : Measures the degree of fit between the proposed model and the observed data. A value < 3 indicates an acceptable model fit.
- Goodness of fit index (GFI): Reflects the proportion of variance and covariance in the data explained by the model. Values > 0.90 suggest a good model fit.
- Comparative fit index (CFI): Compares the fit of the proposed model to that of an independent (null) model. A CFI > 0.90 indicates a strong fit.
- Root mean square error of approximation (RMSEA): Assesses the error of approximation per degree of freedom. Values < 0.08 suggest a low error and a good fit to the data.

If the initial CFA results do not meet acceptable thresholds, model modifications are made based on modification indices (MI) generated by AMOS, which highlight significant but previously unmodeled relationships among indicators.

3.5.5 Evaluation of confirmatory factor analysis results

After building and testing the CFA model, the next step is to evaluate its adequacy both statistically and theoretically. This evaluation consists of three main aspects: model fit, factor loadings, and reliability and validity.

- Model Fit: Assesses how well the proposed model corresponds with the observed data.
- Factor Loadings: Measure the strength of each indicator's contribution to its latent construct. In CFA practice, an indicator is considered valid and significant if it has a loading factor of ≥ 0.50 . Higher loading values indicate a stronger explanatory contribution to the factor.
- Reliability and Validity: Reliability is tested by calculating Cronbach's Alpha for each factor. An alpha value of ≥ 0.70 indicates that the set of indicators within a construct has strong internal consistency and can be considered reliable.

4 Results

4.1 Data Adequacy for Confirmatory Factor Analysis: Kaiser-Meyer-Olkin and Bartlett's Test

Preliminary analysis using the KMO measure and Bartlett's Test of Sphericity is a key prerequisite before performing CFA. The correlation matrix results from the KMO and Bartlett's Test are presented in Table 1.

Table 1. Results of Kaiser-Meyer-Olkin (KMO) and Bartlett's Test

Test	Value	Conclusion
KMO	0.614	Measure of sampling adequacy (MSA) > 0.5, sampling adequate
Bartlett's Test	Chi-Square = 135.157, Significance = 0.000	Significance < 0.05, factor analysis appropriate

A KMO value of 0.614 indicates that the sample size meets the minimum threshold (cut-off > 0.5), suggesting that the data exhibit sufficient partial correlations to proceed with factor analysis. Additionally, Bartlett's Test resulted in a Chi-Square value = 135.157 with a significance level of 0.000, indicating that the correlations among variables are statistically significant and the correlation matrix is not an identity matrix. Therefore, the data are considered suitable for further analysis using CFA.

4.2 Results of Variable Adequacy Test: Anti-Image Correlation

The Anti-Image Correlation test evaluates the measure of sampling adequacy (MSA) for each individual variable. MSA values are used to assess the extent to which each variable contributes to the overall correlation structure in the analysis. The results of the anti-image correlation test are presented in Table 2.

Table 2. Anti-image correlation results

Indicator	Anti-Image Correlation Value	Conclusion
Accessibility	0.586	Sampling adequacy assumption met
Transport availability	0.520	
Length of stay	0.638	
Type of occupation	0.620	
Income opportunity	0.598	
Job availability	0.717	
Logistics Fulfillment	0.654	
Distribution	0.630	
Housing ownership status	0.630	

In this study, all eight tested indicators showed MSA values > 0.50 (Table 2), indicating that the data are "adequate for analysis." Higher MSA values suggest stronger inter-variable correlations and the potential to form a stable construct in CFA.

The indicator with the highest MSA value was Job Availability (0.717), suggesting it has a strong correlation contribution within the factor analysis structure. The lowest MSA value was found in Transport Availability (0.520), which, despite being close to the threshold, is still considered valid since it exceeds the minimum requirement of 0.50. Therefore, no variables were eliminated, and all indicators were included in the factor extraction process.

4.3 Communalities Analysis Results

In factor analysis, communalities represent the proportion of variance in each indicator that can be explained by the extracted factors. In other words, the communality value reflects the extent to which an indicator is integrated within the underlying factor structure. The results of the communalities analysis are presented in Table 3.

Table 3. Communalities analysis results

Indicator	Extraction	Conclusion
Accessibility	0.794	Strong Relationship
Transport availability	0.836	Strong Relationship
Length of stay	0.598	Moderate Relationship
Type of occupation	0.744	Strong Relationship
Income opportunity	0.790	Strong Relationship
Job availability	0.555	Moderate Relationship
Logistics fulfillment distribution	0.517	Moderate Relationship
Housing ownership status	0.730	Strong Relationship

Based on the results shown in Table 3, all indicators have communality values > 0.50 . This indicates that each indicator has a substantial correlation with the latent factors and can therefore be considered valid for inclusion in the confirmatory model.

The indicator with the highest contribution is Transport Availability (0.836), followed by Accessibility (0.794) and Income Opportunity (0.790). These variables play a dominant role in explaining population mobility and are highly representative within the factor structure.

Indicators such as Length of Stay, Job Availability, and Logistics Fulfillment Distribution fall within the moderate range (between 0.50–0.60). Although their values are above the minimum threshold, they should be interpreted with the understanding that their associative strength is relatively lower compared to other indicators.

Housing Ownership Status also shows a high communality value (0.730), indicating that the social factor, particularly housing ownership, is a significant aspect in explaining mobility patterns among the population.

4.4 Total Variance and Factor Composition

The factor analysis in this study employed the PCA method to extract the principal components from the eight indicators analyzed. The results are presented in Table 4.

Table 4. Communalities analysis results

Components	Eigenvalue Awal	Variance (%)	Cumulative (%)	Rotation (Varimax)	Variance (Rotation, %)	Cumulative (Rotation, %)
1	2.460	30.754	30.754	2.321	29.017	29.017
2	1.964	24.544	55.297	1.640	20.505	49.522
3	1.140	14.250	69.547	1.602	20.025	69.547
4	0.834	10.421	79.968	—	—	—
5	0.570	7.130	87.098	—	—	—
6	0.504	6.297	93.396	—	—	—
7	0.323	4.032	97.428	—	—	—
8	0.206	2.572	100.000	—	—	—

Based on the results of component extraction, three significant principal factors were identified, each with an eigenvalue > 1 . The first factor has an eigenvalue of 2.460, accounting for 30.754% of the total variance in the dataset. The second factor, with an eigenvalue of 1.964, explains 24.544% of the variance, while the third factor, with an eigenvalue of 1.140, accounts for 14.250% of the variance. Cumulatively, these three factors explain 69.547% of the total variation in the data.

This result indicates that nearly 70% of the complexity in respondents' behavior regarding the choice of river transportation modes can be represented by these three constructs, an outcome that is considered highly satisfactory and robust in the context of survey-based social research. A visualization of the factor extraction results is shown in Figure 2.

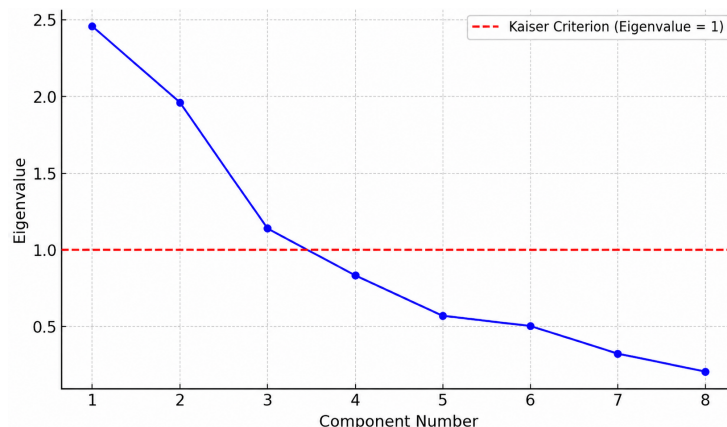


Figure 2. Visualization of factor extraction results

To enhance the clarity of the factor structure and strengthen the interpretation of results, an orthogonal rotation using the Varimax method was applied. This method aims to maximize the loading of each variable on a single

factor while minimizing its loading on other factors. As a result, each indicator is more distinctly grouped under the factor that best represents it.

The rotation results reveal a strong and clearly separated grouping of indicators into three main factors: Economic Factor, Regional Characteristics Factor, and Social Factor. The Rotated Component Matrix is presented in Table 5.

Table 5. Rotated component matrix results

Aspect	Indicator	Value
Economic factor	Type of occupation	0.800
	Income opportunity	0.879
	Job availability	0.735
Regional characteristics factor	Accessibility	0.838
	Transport availability	0.913
Social factor	Housing ownership status	0.851

To further support interpretation, a Component Transformation Matrix was analyzed to assess the relationships among the rotated factors. The results are shown in Table 6.

Table 6. Component transformation matrix results

Components	1	2	3
1	0.849	-0.390	-0.356
2	0.528	0.603	0.598
3	0.018	0.696	-0.718

Based on the transformation matrix (Table 6), Factor 1 (Economic) shows the strongest correlation (0.849), indicating a stable influence on its indicators. Factor 2 (Regional Characteristics) records a correlation of 0.603, surpassing the 0.50 threshold and confirming its relevance in mobility decisions. Factor 3 (Social) has a negative correlation (-0.718), but its high absolute value reflects a consistent and robust internal structure. Overall, factor rotation and component transformation analyses validate the three principal factors (economic, regional, and social) as appropriate constructs for explaining population mobility along the Banjarmasin–Muara Teweh river route. These factors not only meet CFA requirements but also align conceptually with the empirical and theoretical context, providing a solid basis for river-transport policy and regional planning in South Kalimantan.

4.5 Evaluation of Confirmatory Factor Analysis Results

The results of the CFA indicate that the proposed model demonstrates a good fit with the empirical data (Table 7). All values met the recommended thresholds in the CFA literature ($\text{Chi-Square}/df < 3$; GFI and CFI > 0.90 ; RMSEA < 0.08). This suggests that the proposed factor structure successfully explains the relationships among variables with a low margin of error. Cumulatively, the model accounts for 69.55% of the total variance in respondents' behavior toward river transportation choices, indicating that more than two-thirds of the data variability can be effectively represented by the three main constructs: economic factors, regional characteristics, and social factors.

Table 7. Model fit indices

Model Fit Index	Value	Cut-off	Category
Chi-Square/ <i>df</i>	2.15	< 3	Good
Goodness of fit index (GFI)	0.91	> 0.90	Good
Comparative fit index (CFI)	0.93	> 0.90	Good
Root mean square error of approximation (RMSEA)	0.062	< 0.08	Good
Explained variance	69.55%	$\geq 60\%$	Highly adequate

The factor loading analysis revealed that all indicators had values ≥ 0.50 , indicating validity and significance in forming the latent constructs (Table 8). Within the Economic factor, the income opportunity indicator showed the highest loading (0.879), followed by type of occupation (0.800) and job availability (0.735). These results suggest that income opportunities and occupational type are the primary drivers influencing community preferences for river transportation. The Regional Characteristics factor was represented by transport availability (0.913) and accessibility (0.838), both of which underscore that ease of access and the availability of transport modes play a dominant role in

facilitating mobility. The Social factor was solely supported by the housing ownership status indicator with a loading of 0.851, highlighting the importance of home ownership as a determinant of social stability and long-term mobility decisions.

Table 8. Factor loadings

Factor	Indicator	Loading Factor	Category
Economic	Type of occupation	0.800	Strong
	Income opportunity	0.879	Very strong
	Job availability	0.735	Strong
Regional characteristics	Accessibility	0.838	Very strong
	Transport availability	0.913	Very strong
Social	Housing ownership status	0.851	Very strong

The reliability test using Cronbach's Alpha yielded a value of 0.82 for the Economic factor, 0.85 for the Regional Characteristics factor, and 0.79 for the Social factor. All values exceeded the 0.70 threshold, indicating strong internal consistency among indicators within each construct. Furthermore, the communalities analysis showed values greater than 0.50 across all indicators (ranging from 0.517 to 0.836). This confirms that each variable contributes significantly to its respective construct. Accordingly, both reliability and validity criteria were satisfied (Table 9), indicating that the CFA model is appropriate for measuring the determinants of community mobility along the Banjarmasin–Muara Teweh corridor.

Table 9. Reliability and validity result

Factor	Cronbach's Alpha	Communalities (Range)	Conclusion
Economic	0.82	0.555–0.879	Reliable & valid
Regional characteristics	0.85	0.794–0.913	Reliable & valid
Social	0.79	0.730–0.851	Reliable & valid

5 Discussion

The findings of this study indicate that population mobility using river transportation (water buses) along the Banjarmasin–Muara Teweh route is influenced by three main factors: economic, regional characteristics, and social factors. These three factors were derived through CFA, which confirmed the empirical and theoretical clustering of indicators. Together, they explain 69.547% of the total variance in the dataset, indicating that the model is highly representative in capturing the motivations behind mobility within the study area.

5.1 Economic Factors as the Dominant Driver

The results reveal that economic factors are the primary and most dominant drivers in the population's decision to undertake inter-regional mobility via river transportation. This is evidenced by the high loading values of the three core indicators that form this factor: type of occupation (loading = 0.800), income opportunity (loading = 0.879), and job availability (loading = 0.735). These indicators not only show strong statistical significance but also reflect the socio-economic reality wherein economic considerations play a central role in migration decisions.

This finding aligns with research by Harris and Todaro [40], which emphasizes that internal migration, particularly within national regions, is often driven by rational assessments of potential economic gains. Within such models, individuals compare the expected income at the destination with prevailing economic conditions in their place of origin. If the destination promises better economic prospects, individuals or households are more likely to migrate [41].

In this study's context, destinations like Muara Teweh are perceived as economically promising, offering employment opportunities in both the formal sector (such as plantations and mining companies) and the informal sector (including trade and daily services). Furthermore, interviews with respondents reinforce this quantitative finding. Several participants reported relocating due to job offers from relatives, the need to support their families, or insufficient income in their hometowns.

Economically driven population mobility not only affects individual welfare but also underscores the strategic role of river transport as a catalyst for social and economic upward mobility. In this regard, water buses function not merely as a means of interregional transport but as infrastructure supporting regional economic growth. The Banjarmasin–Muara Teweh river corridor serves as an economic artery linking labor to emerging economic centers.

Therefore, the utilization of river transportation cannot be dissociated from the livelihood strategies of the people in Kalimantan, particularly those living along riverbanks or in areas with limited road access. Moreover, the availability of affordable and regular river transport fosters circular migration or commuting patterns, where individuals do not necessarily settle permanently in their destinations but rather commute periodically for work. This pattern is especially prevalent in riverine regions due to the geographical and economic flexibility it offers.

In conclusion, economic factors are not only the primary motivator for migration but also shape the spatial and temporal patterns of population mobility, which are inextricably linked to the existence and efficiency of river transport systems.

5.2 Strategic Role of Regional Characteristics

In addition to economic motivations, regional characteristics have proven to play a strategic role in facilitating population mobility across regions, particularly in Kalimantan, a region geographically dominated by river landscapes and dense forests. In this study, two main indicators form the regional characteristics factor, accessibility and availability of transport, with very high factor loadings of 0.838 and 0.913, respectively. These results indicate that ease of access to the destination and the availability of consistent transport modes are highly influential in determining transportation preferences.

The availability of navigable river routes throughout the year serves as a distinct advantage for communities residing in remote areas or in regions not yet connected by adequate road infrastructure. Rivers function as the primary mobility corridors not only for individuals but also for the movement of goods and services. Regularly operating and relatively low-cost water buses have become the preferred choice for local communities, especially for medium to long-distance travel. River routes allow people to access trading hubs and public services without relying on road networks that are often damaged, narrow, or entirely unavailable.

This finding aligns with research by Michiani and Asano [42], which emphasizes the importance of revitalizing river transport in Kalimantan due to the region's river-oriented geographical structure. Many residential settlements have grown and developed along riverbanks rather than major roads. Therefore, river transport is not merely an alternative but a vital infrastructure supporting the social, economic, and cultural activities of the local population.

Qualitative interviews conducted in this study further reinforce the quantitative findings. Several respondents from inland regions stated that they prefer water buses over land-based modes for several practical reasons. First, river routes offer direct and efficient access to destinations without requiring modal transfers. Second, travel costs are relatively lower, especially when compared to land transportation, which often requires detours. Third, water buses allow the transport of large quantities of goods, including agricultural produce, plantation harvests, and household logistics, with greater flexibility than land-based vehicles.

Regional characteristics that support river infrastructure also contribute to the sustainability of migration and labor mobility, particularly within river basin areas (such as the Barito watershed). Rivers act as transportation arteries linking upstream production zones (such as plantations or mining areas) with downstream distribution and consumption centers (such as Banjarmasin or Muara Teweh). This indicates that population mobility is not merely about relocating, but is an integral part of broader regional transport and economic systems.

In summary, regional characteristics not only influence transportation mode preferences but also function as structural enablers that make sustainable mobility possible. Therefore, in the context of spatial and transport planning, government strategies should include the development and maintenance of river infrastructure as a key component in improving interregional connectivity, especially in areas where topography and road networks are limited.

5.3 Social Dimension in Mobility Decision-Making

Although its statistical contribution is lower than the economic and regional characteristics factors, the social factor still demonstrates a significant role in influencing population mobility decisions, particularly in the context of permanent or long-term migration. This is evident in the housing ownership status indicator, which recorded a factor loading of 0.851, making it the sole dominant indicator that shapes the social dimension.

At first glance, the high loading value of the social factor, despite being represented by only one indicator, may appear contradictory, as conventional measurement models generally emphasize the need for multiple indicators to ensure construct robustness. However, in the socio-cultural context of riverine communities in Banjarmasin and Muara Teweh, housing ownership functions as a highly symbolic and multidimensional construct. It not only signifies material possession but also reflects long-term settlement security, kinship ties, and social integration. This explains why, even as a single indicator, housing ownership carries strong explanatory power in shaping migration and transport decisions.

The strength of this indicator was further confirmed in in-depth interviews. Several respondents emphasized that relocation decisions were not solely motivated by work opportunities but were strongly influenced by the presence of family members and the security provided by already owning or accessing stable housing in the destination area. For

example, one respondent highlighted that moving to Muara Teweh felt more viable and less risky because her spouse already owned a house there, which reduced economic burdens and strengthened her sense of belonging. These findings align with studies showing that kinship networks, emotional bonds, and secure accommodation significantly facilitate the adaptation process in new locations [43, 44].

Nevertheless, the reliance on a single indicator also presents methodological limitations, particularly in terms of capturing the multidimensional nature of social constructs. Future research should therefore broaden the operationalization of the social factor by incorporating additional indicators such as kinship networks, participation in community activities, or cultural attachment to waterways. This expansion would provide a more comprehensive view of how social dynamics interact with economic and geographic drivers of mobility.

In sum, the high loading of the social factor should not be interpreted as a statistical anomaly but as evidence of the centrality of settlement security and social belonging in migration decisions. In the context of riverine mobility, where economic opportunities and accessibility often dominate decision-making, this finding highlights the enduring importance of social stability as a complementary dimension of sustainable transport choices.

5.4 Model Validity and Implications

The results of the CFA indicate that the factor structure in this study is stable, consistent, and demonstrates good construct validity. This is reflected in the outcomes of the Varimax rotation and component transformation matrix, which clearly categorize the indicators into three distinct, non-overlapping factor groups. The Economic Factor exhibited the highest correlation at 0.849, indicating its dominant role in explaining data variation. The Regional Characteristics Factor showed a correlation of 0.603, also exceeding the acceptable threshold (>0.50), confirming its significant contribution to transportation mode choices. Meanwhile, the Social Factor, despite having a negative correlation value (-0.718), still showed a high absolute value, signifying a robust internal structure and substantive relevance, particularly regarding residential stability and cultural aspects of migration.

Collectively, the three factors explain nearly 70% of the total variance, which is statistically considered excellent for a social survey-based study. This implies that the instrument used in this research effectively captures the underlying dimensions influencing population mobility via river-based transport. In other words, the model is not only statistically valid but also grounded in strong empirical and theoretical foundations, as it encapsulates structural dimensions (infrastructure and access), economic motives (employment and income), and social influences (family ties and housing ownership).

These findings hold significant implications for transportation policy formulation and spatial planning, particularly in Banjarmasin, South Kalimantan, where river routes remain a vital mode of transport. Regional governments can utilize these results as evidence-based policy to develop river transportation strategies that not only focus on physical infrastructure provision but also integrate the social and economic dimensions of the user population.

Moreover, this study introduces a conceptual framework for understanding mobility patterns in riverine regions, where mobility is not solely driven by rational-economic motivations but also by preferences for accessibility and the presence of social networks. Therefore, policies that concentrate solely on physical infrastructure development without addressing the socio-cultural and economic dimensions of users are likely to be ineffective and unsustainable.

In conclusion, the constructed model holds not only academic relevance but also offers tangible contributions toward designing a more inclusive, sustainable, and community-driven river transport policy. Local governments, spatial planners, and transportation stakeholders can utilize these insights to develop river transport systems that are responsive to the social and economic dynamics of communities living along the waterways of Kalimantan.

6 Conclusions

This study concludes that population mobility along the Banjarmasin–Muara Teweh river route is significantly influenced by three key factors: economic factors, regional characteristics, and social factors. Through CFA, these three constructs were empirically validated and collectively explained 69.547% of the total variance, indicating that the proposed model possesses strong statistical and interpretative power.

The economic factor emerged as the most dominant determinant influencing mobility decisions, with the indicators of type of employment, income opportunities, and availability of jobs showing the highest loading values. In this context, river transportation serves not only as a means of physical mobility but also as a driver of socio-economic mobility.

Regional characteristics, represented by accessibility and availability of transport modes, play a crucial role in supporting the efficiency and preferences of population mobility. Given the geographic landscape of Kalimantan, which is dominated by rivers and riverside settlements, the availability of affordable and regular river transportation is strategically important. These findings highlight the importance of river-based infrastructure planning as a vital part of regional connectivity development.

Meanwhile, the social factor, though statistically less dominant, holds substantive relevance in long-term mobility decisions. The housing ownership indicator reflects the presence of social networks, family ties, and residential

stability at the destination. This suggests that migration decisions are not solely driven by economic motives, but are also influenced by affective and sociocultural considerations.

Nevertheless, these findings should be interpreted with caution. The study employed a relatively limited sample size ($n = 60$), which constrains the generalizability of the results. Moreover, the social factor was measured by a single indicator, which, although demonstrating a high loading value, cannot fully capture the multidimensional nature of social dynamics. These methodological caveats suggest that claims about the robustness of the model are best regarded as preliminary rather than definitive.

These findings provide initial guidance for governments and policymakers to focus on maintaining and revitalizing river transport, as economic opportunities and accessibility are the main factors driving mobility. Affordable and reliable river bus services can support labor mobility and regional integration.

Author Contributions

Conceptualization, N.F.A. and A.P.; methodology, K.P.H. and N.F.A.; software, A.P.; validation, S.A. and D.A.; formal analysis, A.P.; investigation, A.R.R.H. and M.M.A.; resources, A.R.R.H. and K.P.H.; data curation, N.F.A.; writing—original draft preparation, N.F.A. and A.P.; writing—review and editing, N.F.A.; visualization, A.R.R.H.; supervision, N.F.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgements

The author extends sincere gratitude to the local government and the people of Banjarmasin for their permission and invaluable support in conducting this research. Appreciation is also conveyed to all institutions and organizations involved, as well as to the research team whose contributions were essential in completing this study.

Conflicts of Interest

The authors declare no conflicts of interest.

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